

Heat-Kernel Cesàro Limits and Dixmier Measurability: a Later Literature Answer

Status note for arXiv:0912.2817

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Status: literature_already_answered. This note records a separate later-paper answer; it is not an original solution produced by the run.

Source question

Carey, Gayral, Rennie, and Sukochev, *Integration on locally compact noncommutative spaces*, arXiv:0912.2817 [1], define for $T \in \mathcal{Z}_1$

$$f_T(\lambda) = \lambda^{-1} \tau \left(e^{-\lambda^{-1} |T|^{-1}} \right)$$

and use the multiplicative Cesàro operator

$$(Mh)(\lambda) = \frac{1}{\log \lambda} \int_1^\lambda h(t) \frac{dt}{t}.$$

On source PDF page 9, after discussing independence of Dixmier traces from the generalized limit, they state that existence of

$$\lim_{\lambda \rightarrow \infty} (Mf_T)(\lambda) \tag{1}$$

is sufficient for measurability and ask whether it is necessary.

Supporting answer

Carey and Sukochev, *Measurable operators and the asymptotics of heat kernels and zeta functions*, arXiv:1201.3661 [2], explicitly announce that they answer the heat-kernel measurability questions considered there. Their Theorem 7, on supporting PDF page 5, states that for every positive $A \in \mathcal{M}_{1,\infty}$, the following are equivalent:

1. A is measurable for the class of Dixmier traces;
2. the logarithmic singular-value mean has an ordinary limit;
- 3.

$$\lim_{\lambda \rightarrow \infty} M \left(\lambda \mapsto \lambda^{-1} \tau(e^{-(\lambda A)^{-1}}) \right)$$

exists;

4. $\lim_{s \rightarrow 0} s \tau(A^{1+s})$ exists.

The theorem also proves that the three limits and the common Dixmier-trace value coincide. Taking $A = |T|$, condition (iii) is exactly (1). Thus the requested necessity holds, and in fact belongs to a stronger necessary-and-sufficient four-way characterization.

Scope and provenance

The source question is about the positive heat-kernel quantity attached to $|T|$; the supporting theorem applies to positive operators in $\mathcal{M}_{1,\infty}$, the same Marcinkiewicz ideal denoted $\mathcal{Z}_1$ in the source. The answer paper is separate and later, and two of its authors are authors of the source paper. Its abstract and introduction say that it answers the outstanding measurability/heat-kernel questions, and Theorem 7 gives the exact equivalence.

The queue’s earlier natural-question signal on the same source page asks whether f_T may be unbounded while Mf_T is bounded. That question is answered immediately by the following proposition in arXiv:0912.2817 and is therefore a same-paper extraction false positive. This status packet records only the distinct genuine question above.

References

- [1] A. L. Carey, V. Gayral, A. Rennie, and F. A. Sukochev, *Integration on locally compact noncommutative spaces*, J. Funct. Anal. **260** (2011), 1116–1163; arXiv:0912.2817.
- [2] A. Carey and F. Sukochev, *Measurable operators and the asymptotics of heat kernels and zeta functions*, J. Funct. Anal. **262** (2012), 4589–4599; arXiv:1201.3661.